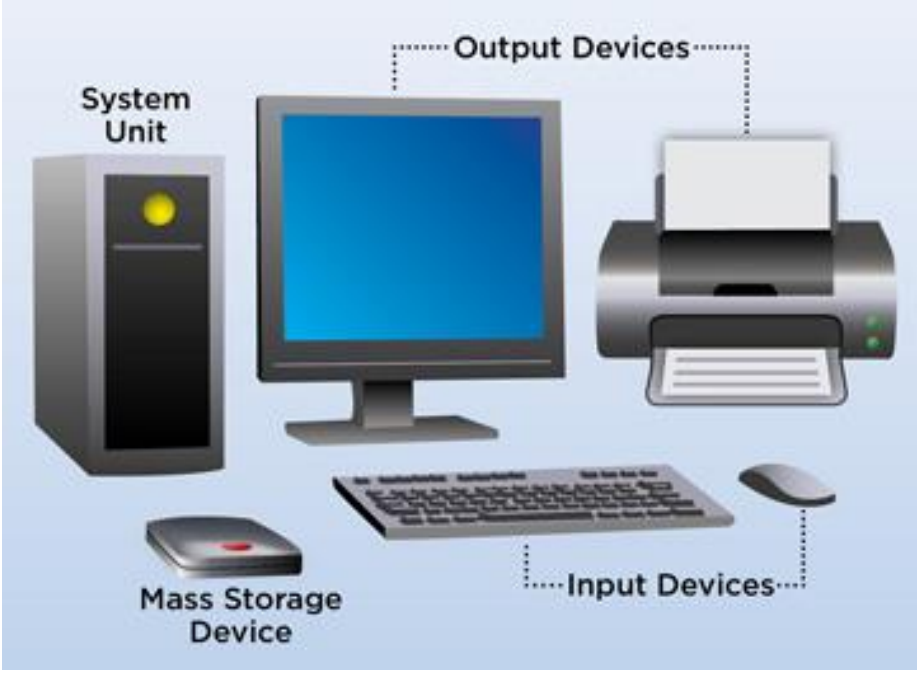
Module 1: Basic data operations

Fundamentals of computations

Hardware



Software



Fundamentals of computations

Computers generally operate in a 2 step cycle

1. Fetch the instruction
2. Execute the instruction

These two steps are repeated over and again.

So, computers are basically a combination of Information processing system (CPU, memory) and environment (Input and output devices).

Instruction or set of instructions processed by the CPU is called a program.

Text form of these instruction is called a CODE.

Fundamentals of computations

CPU can only process the instruction that are in machine language.

Machine language, or machine code is comprised of binary digits (0 and 1).

CODE developed by the humans are in text from and not binary. So, to process these instructions, text CODE must be converted into the binary form.

This can be achieved via

1. Compiling
2. Interpreting

Fundamentals of computations

Compilation: It is the process of converting text CODE into machine code. Here, entire set of instruction are first converted into machine code. This can later be executed to achieve the desired result.

Example: C/C++

Interpretation: Interpreters simultaneously convert and execute the instructions of the CODE. This is done in a line-by-line manner.

Example: Python

Compiled CODEs are generally faster. However, they have platform dependence of the generate machine code.

Interpreter CODEs are generally more flexible and dynamic, but are usually slower.

Fundamentals of computations

A programming language generally has following components

1. **Data type**: Object defined in the program will have some basic characteristic

Example: real or integer.

2. **Data structure**: Data can be represented in complex form

Example: Arrays.

3. **Variables**: Place to store the data or values

4. **Assignments**: It is a statement that sets and/or re-sets the value stored in the variable

Example: a=3+1.

Fundamentals of computations

5. **Function**: It is a command that accomplish a specific task. It generally takes an argument and return a value

Example: print “hello world” **--->** hello world

6. **Control structure**: Control structures are specific commands that dictate which part of the program the computer will executed next.

Example: **If** command

Data visualization

Data

Data is/are a set of values of qualitative or quantitative variables.

Data in itself has a little meaning.

Information

Information is obtained via post-processing of data.

Information has a specific meaning to it.

Information is specific to the requirement.

Data visualization

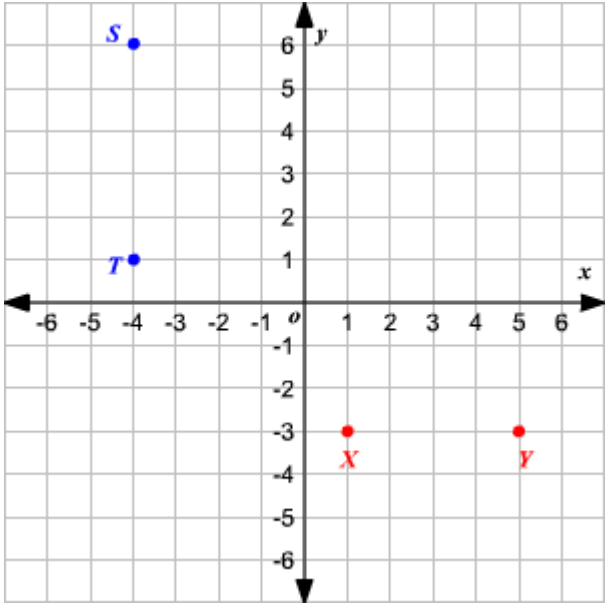
Plotting

A plot is a graphical technique for representing a data set, exhibiting the relationship between two or more variables.

They are an important post processing method.

Data sets can graphically be represented in various forms

Line chart
Pie chart
Log plot



Data visualization

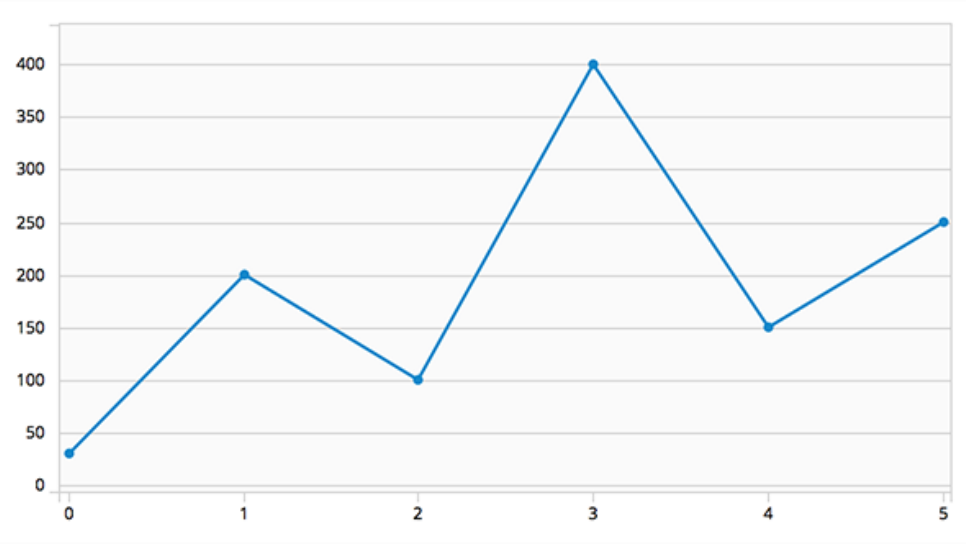
Line chart

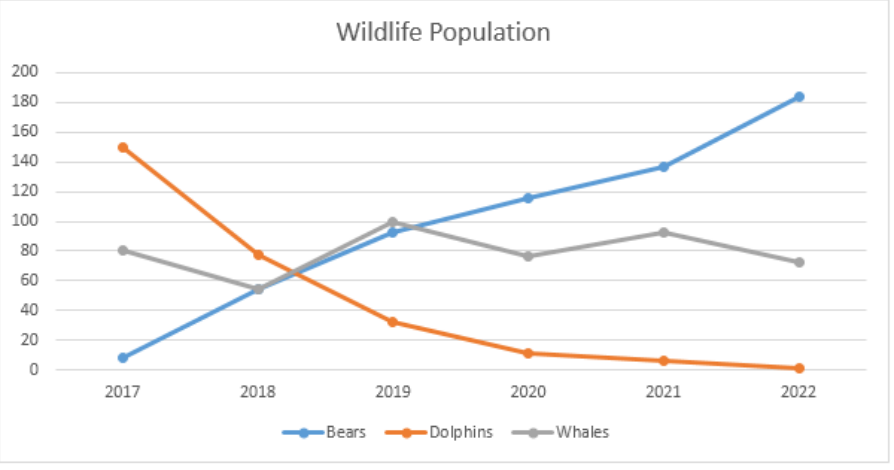
Also known as line plot or curve plot or line graph.

This displays information as a series of data points called 'markers'.

A line chart is often used to visualize a trend in data.

A multi-line chart can be used to compare the behavior of same variables in different environments or conditions.

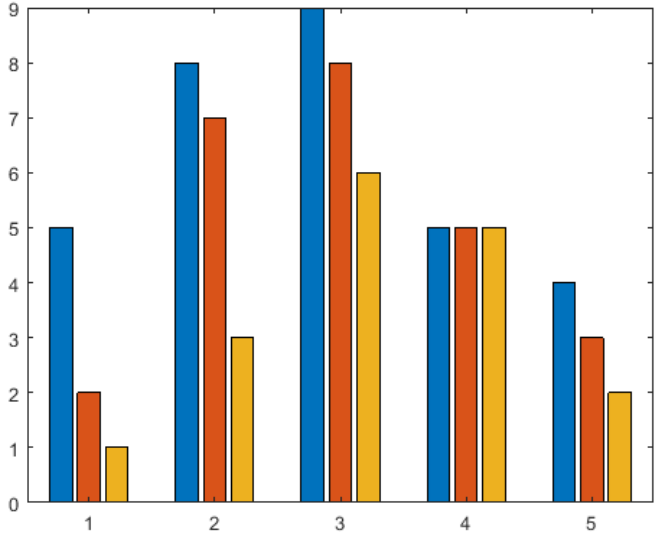




Data visualization

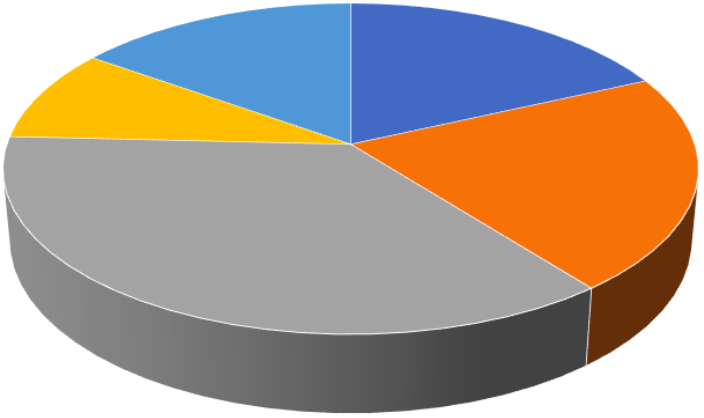
Bar Graphs

Bar graph or bar chart shows the comparison between the discreet categories.



Pie Chart

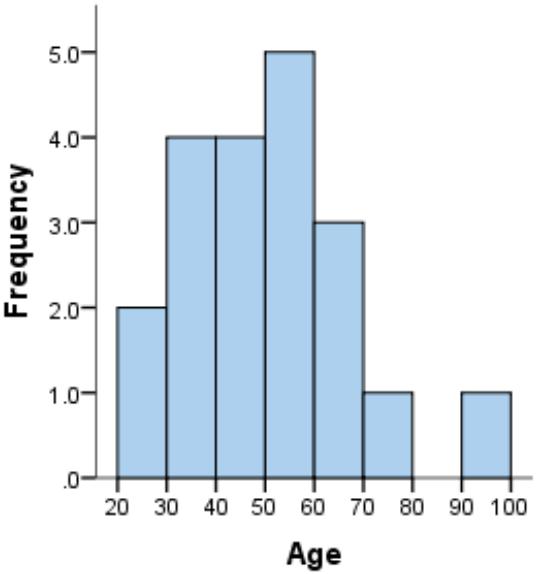
It is a circular statistical graph, with each slice representing the numerical proportion of the category.



Data visualization

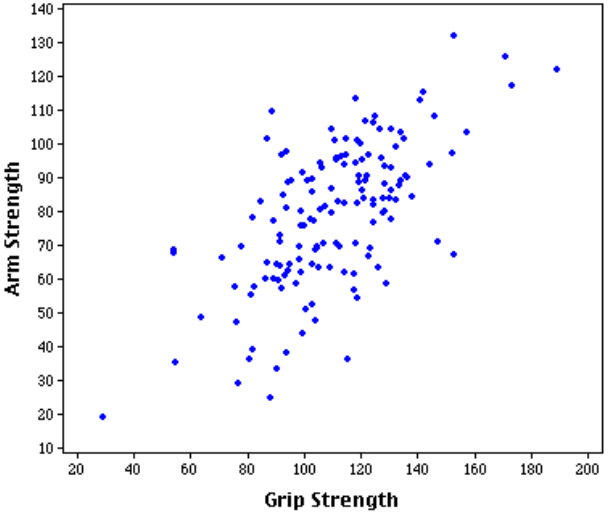
Histogram

Histogram represent the distribution of a numerical data. X-axis and the Y-axis represent the range and the frequencies respectively.



Scatter plots

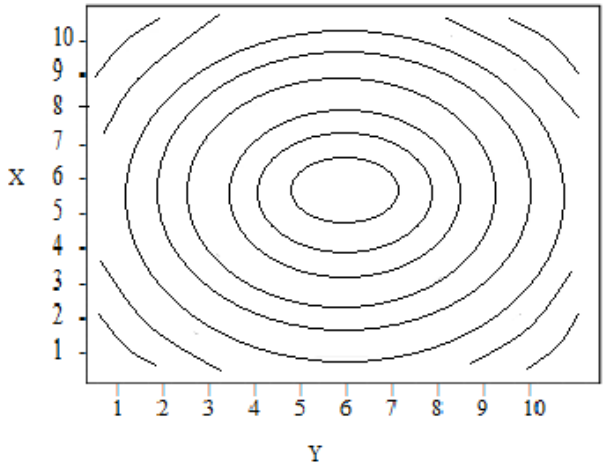
A scatter plot uses dots to represent values for two different numeric variables. If the points are color codes can be used here to display one additional variable.



Data visualization

Contour plot

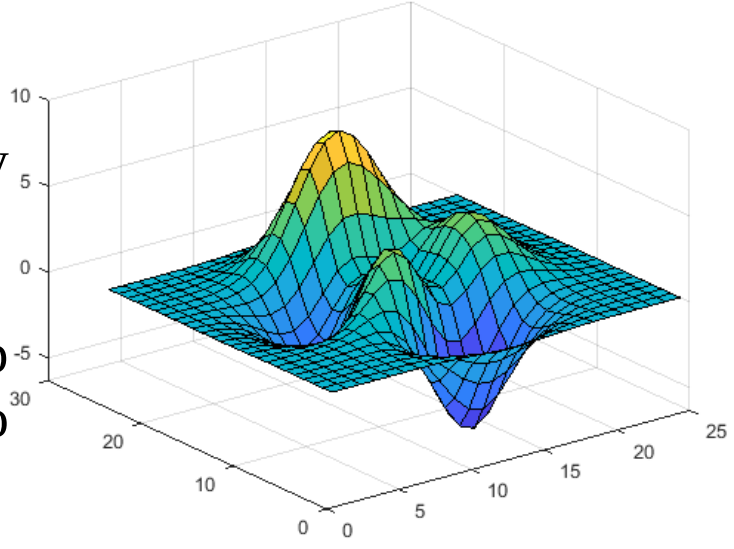
A contour plot is a graphical technique for representing a 3-dimensional surface by plotting constant z slices, called contours, on a 2-dimensional format.



3-D plots

It is the three dimensional representation of data. They are also known as surface plot.

Here, individual data points show the relationship between designated dependent variable (Y), and two independent variables (X and Z).



Data interpretation

Data interpretation is the process of reviewing data through some predefined processes which will help assign some meaning to the data and arrive at a relevant conclusion.

It involves taking the result of data analysis, making inferences on the relations studied, and using them to conclude.

Data analysis is the process of ordering, categorizing, manipulating, and summarizing data to obtain answers to research questions. It is usually the first step taken towards data interpretation.

There are two main methods of data interpretation

1. Qualitative methods
2. Quantitative methods

Data interpretation

Qualitative methods

Qualitative data, also known as categorical data, is a collection of information that is divided into groups.

Qualitative data can take on numerical values, but those numbers have little mathematical meaning.

Qualitative data can be classified into two categories:

1. **Nominal data**: This is a type of data used to name variables without providing any numerical value.

2. **Ordinal data**: This is a data type with a set order or scale to it. However, this order does not have a standard scale on which the difference in variables in each scale is measured.

Data interpretation

Qualitative methods

Qualitative data is analyzed using mean or distribution etc.

Qualitative data analyzed graphically using a bar chart and pie chart.

In most of the cases, qualitative data has to be converted into numerical data before the processing. However, these values do not exhibit quantitative characteristics. Arithmetic operations can not be performed on them.

Data interpretation

Quantitative methods

Quantitative data, also known as numerical data, is always collected in number form.

Quantitative data can be classified into two categories,

1. **Discrete Data**: This is data type that represents countable items. It take on values that can be grouped into a list.

2. **Continuous Data**: It represents a data type that can assume any value. Continuous Data represents measurements and therefore their values can't be counted but they can be measured.

Continuous data can further be divided into two catagories: (a) Interval data, and (b) ratio data

Data interpretation

Quantitative methods

a. **Interval data**: This is a data type measured along a scale, in which each point is placed at an equal distance from one another.

b. **Ratio data**: This data type has same properties as interval data, with an equal and definitive ratio between each data and absolute “zero” being treated as a point of origin. Thus, there can be no negative numerical value in ratio data.

Quantitative data can be analyzed using a variety of methods such as

Descriptive analysis methods: Mean, median mode etc.

Inferential statistical methods: Trend analysis

Data interpretation

Quantitative methods

Quantitative data can be utilized for estimation or enumeration.

One can perform mathematical operation on the quantitative data.

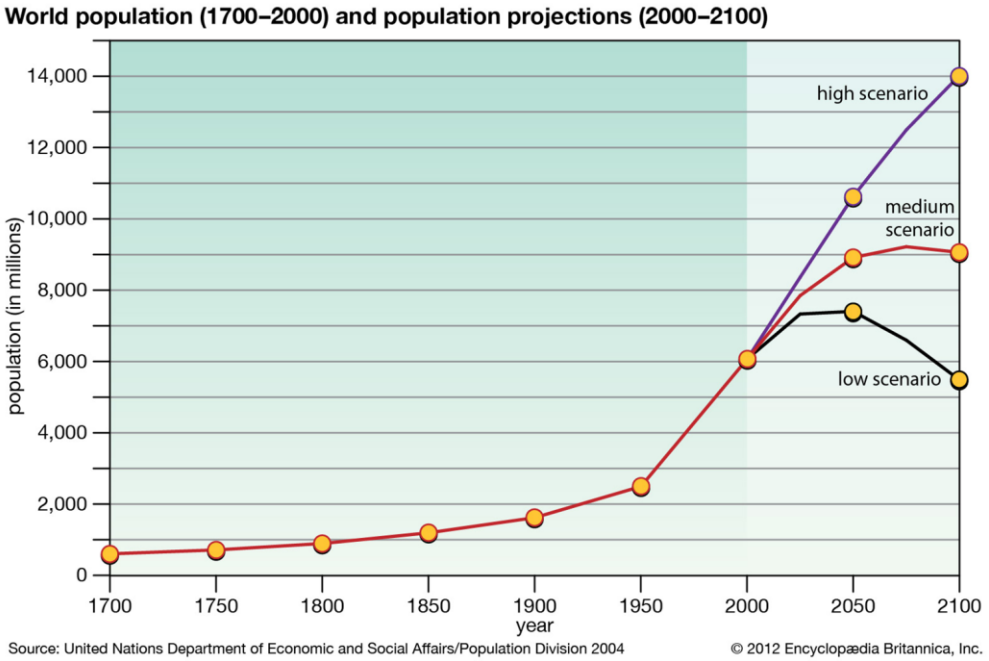
Quantitative data may be visualized in different ways depending on the type of data being investigated, e.g., histogram, scattered plot, or line plot etc.

Note: One of the key factor that influence a data is bias. Presence of nay bias may lead to falsification of results. Thus it important to avoid bias during the data collection and interpretation.

Data fitting and (inter-extra)polation

Interpolation or extrapolation is a type of estimation, a method of constructing new data points within the range of a discrete set of known data points.

To interpolate/extrapolate, we require a function that is most suitable to represent the given data set.



Function required for the interpolation is obtained via the curve fitting.

Curve fitting or data fitting is the process of constructing a curve, or mathematical function, that has the best fit to a series of data points.

Data fitting and interpolation

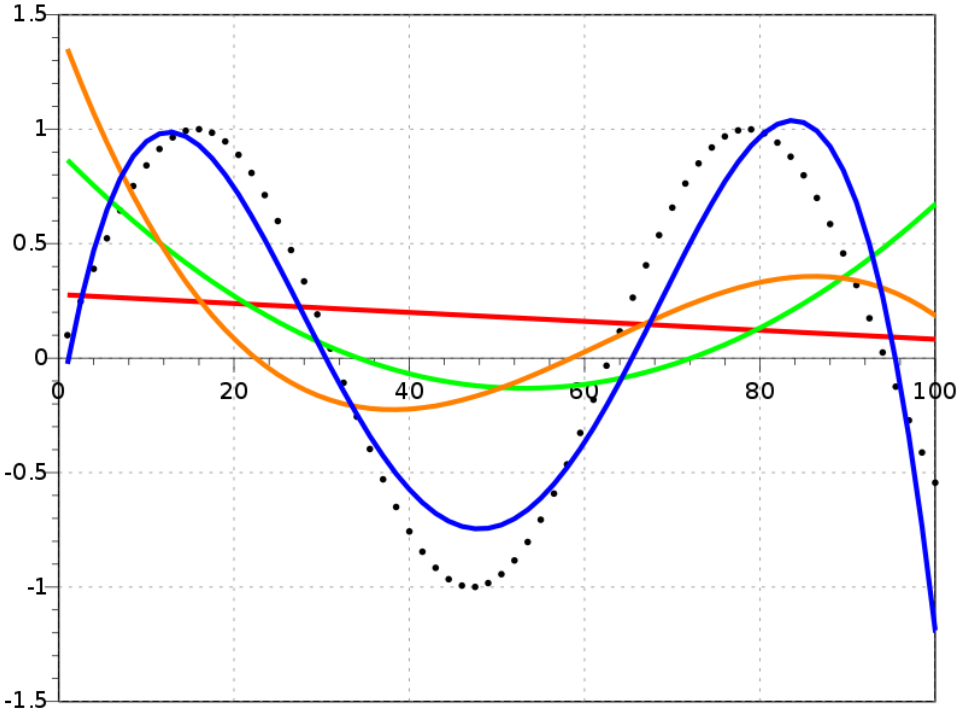
Any data is fitted using the polynomial equation

$y = f(x)$

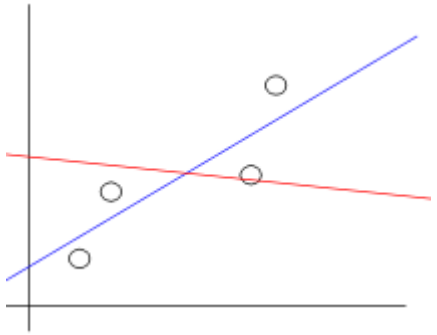
$f(x) = ax + b$ – Linear fitting or first degree polynomial

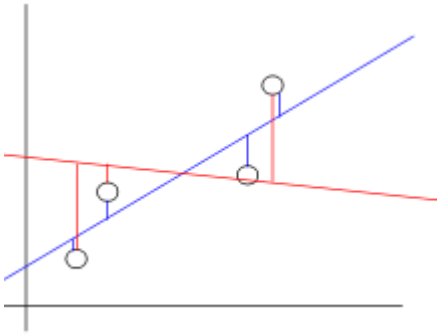
$f(x) = ax + bx^2 + c$ – Second degree polynomial

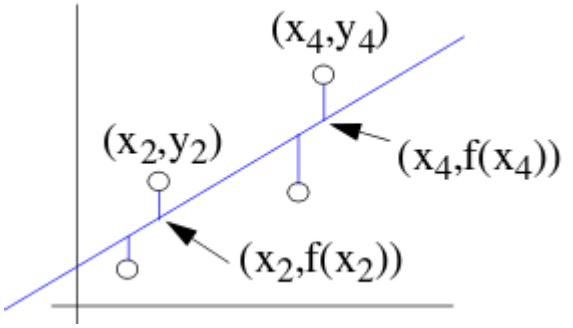
$f(x) = ax + bx^2 + cx^3 + d$ – Third degree polynomial



Curve fitting







Linear fitting

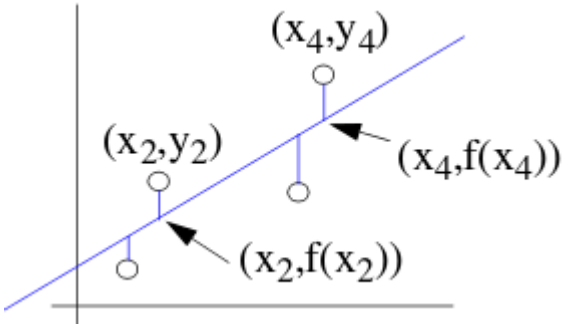
$$f(x) = ax + b$$

$$err = \sum (d_i)^2 = (y_1 - f(x_1))^2 + (y_2 - f(x_2))^2 + (y_3 - f(x_3))^2 + (y_4 - f(x_4))^2$$

What values of ‘a’ and ‘b’ are most appropriate?

$$err = \sum_{i=1}^{\#\text{ data points}} (y_i - f(x_i))^2 = \sum_{i=1}^{\#\text{ data points}} (y_i - (ax_i + b))^2$$

Curve fitting



$$a\sum x_i^2 + b\sum x_i = \sum (x_i y_i)$$
$$a\sum x_i + b*n = \sum y_i$$

$$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} b \\ a \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum (x_i y_i) \end{bmatrix}$$

$$\text{minimize} \quad err = \sum_{i=1}^{\#\text{ data points} = n} (y_i - (ax_i + b))^2$$

$$A = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}, \quad X = \begin{bmatrix} b \\ a \end{bmatrix}, \quad B = \begin{bmatrix} \sum y_i \\ \sum (x_i y_i) \end{bmatrix}$$

$$\frac{\partial err}{\partial a} = -2 \sum_{i=1}^{n} x_i (y_i - ax_i - b) = 0$$

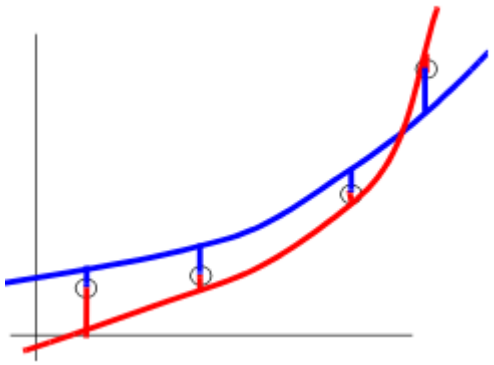
$$AX = B$$

$$\frac{\partial err}{\partial b} = -2 \sum_{i=1}^{n} (y_i - ax_i - b) = 0$$

$$X = A^{-1} * B$$

Curve fitting

Polynomial fitting



$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_j x^j = a_0 + \sum_{k=1}^{j} a_k x^k$$

$a_0, a_1, a_2, a_3, \ldots\ldots\ldots, a_j$?

$$err = \sum (d_i)^2 = (y_1 - f(x_1))^2 + (y_2 - f(x_2))^2 + (y_3 - f(x_3))^2 + (y_4 - f(x_4))^2$$

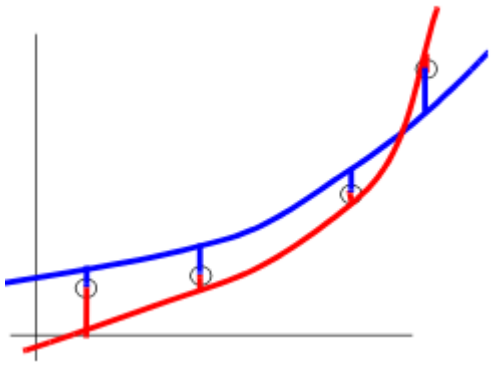
$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_j x^j = a_0 + \sum_{k=1}^{j} a_k x^k$$

$$err = \sum_{i=1}^{n} \left(y_i - \left(a_0 + a_1 x_i + a_2 x_i^2 + a_3 x_i^3 + \ldots + a_j x_i^j \right) \right)^2$$

$$err = \sum_{i=1}^{n} \left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k \right) \right)^2$$

Curve fitting

Polynomial fitting



$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_j x^j = a_0 + \sum_{k=1}^{j} a_k x^k$$

$a_0, a_1, a_2, a_3, \ldots\ldots\ldots, a_j$?

$$err = \sum (d_i)^2 = (y_1 - f(x_1))^2 + (y_2 - f(x_2))^2 + (y_3 - f(x_3))^2 + (y_4 - f(x_4))^2$$

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_j x^j = a_0 + \sum_{k=1}^{j} a_k x^k$$

$$err = \sum_{i=1}^{n} \left(y_i - \left(a_0 + a_1 x_i + a_2 x_i^2 + a_3 x_i^3 + \ldots + a_j x_i^j \right) \right)^2$$

$$err = \sum_{i=1}^{n} \left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k \right) \right)^2$$

Curve fitting

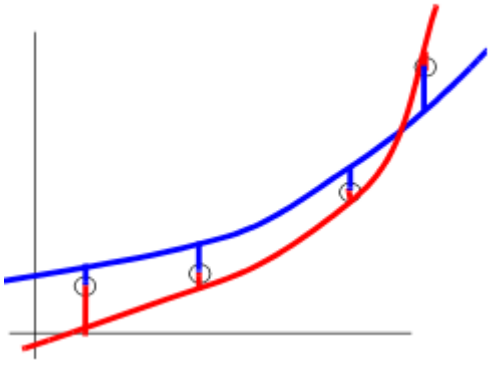
$$\frac{\partial err}{\partial a_0} = -2 \sum_{i=1}^{n} \left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k \right) \right) = 0$$

$$\frac{\partial err}{\partial a_1} = -2 \sum_{i=1}^{n} \left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k \right) \right) x = 0$$

$$\frac{\partial err}{\partial a_2} = -2 \sum_{i=1}^{n} \left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k \right) \right) x^2 = 0$$

$$\vdots$$

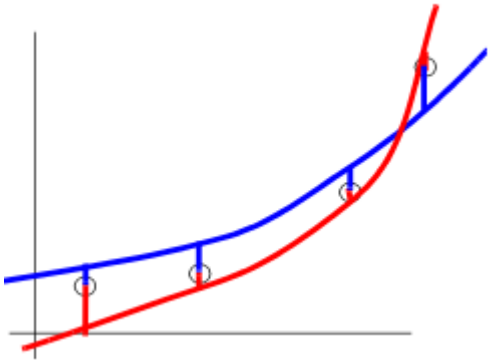
$$\frac{\partial err}{\partial a_j} = -2 \sum_{i=1}^{n} \left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k \right) \right) x^j = 0$$



$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 & \dots & \sum x_i^j \\ \sum x_i & \sum x_i^2 & \sum x_i^3 & \dots & \sum x_i^{j+1} \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 & \dots & \sum x_i^{j+2} \\ \vdots & \vdots & \vdots & & \vdots \\ \sum x_i^j & \sum x_i^{j+1} & \sum x_i^{j+2} & \dots & \sum x_i^{j+j} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_j \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum (x_i y_i) \\ \sum \left(x_i^2 y_i \right) \\ \vdots \\ \sum \left(x_i^j y_i \right) \end{bmatrix}$$

Curve fitting

$$A = \begin{bmatrix} n & \sum x_i & \sum x_i^2 & \dots & \sum x_i^j \\ \sum x_i & \sum x_i^2 & \sum x_i^3 & \dots & \sum x_i^{j+1} \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 & \dots & \sum x_i^{j+2} \\ \vdots & \vdots & \vdots & & \vdots \\ \sum x_i^j & \sum x_i^{j+1} & \sum x_i^{j+2} & \dots & \sum x_i^{j+j} \end{bmatrix}, \quad X = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_j \end{bmatrix}, \quad B = \begin{bmatrix} \sum y_i \\ \sum (x_i y_i) \\ \sum \left(x_i^2 y_i \right) \\ \vdots \\ \sum \left(x_i^j y_i \right) \end{bmatrix}$$



$$AX = B$$

$$X = A^{-1} * B$$

$$na_0 + \Sigma x_i a1 + \Sigma x_i^2 a_2 = \Sigma y_i$$
$$\Sigma X_i a_0 + \Sigma x_i^2 a_1 + \Sigma x_i^3 a_2 = \Sigma x_i y_i$$
$$\Sigma X_i^2 a_0 + \Sigma x_i^3 a_1 + \Sigma x_i^4 a_2 = \Sigma x_i^2 y_{ix}$$

Data plotting - Python

```
import matplotlib.pyplot as plt

x = [1,2,3,4]

y = [1,2,3,4]

plt.plot(x, y)

plt.show()
```

Data plotting - Python

```
import matplotlib.pyplot as plt

x = [1,2,3,4]

y = [1,2,3,4]

plt.plot(x, y)

plt.xlabel('x - axis')

plt.ylabel('y - axis')

plt.show()
```

Data plotting - Python

```
import matplotlib.pyplot as plt

x = [1,2,3,4]

y = [1,2,3,4]

plt.plot(x, y)

plt.xlabel('x – axis')

plt.ylabel('y – axis')

plt.title('My first graph')

plt.show()
```

Data plotting - Python

```
import matplotlib.pyplot as plt

x = [1,2,3,4]

y = [1,2,3,4]

plt.plot(x, y, color='green', linestyle='dashed', linewidth = 3,

        marker='o', markerfacecolor='blue', markersize=12)

plt.xlabel('x – axis')

plt.ylabel('y – axis')

plt.title('My first graph')

plt.show()
```

Data plotting - Python

```
import matplotlib.pyplot as plt

x1 = [1,2,3,4]

y1 = [1,2,3,4]

plt.plot(x1, y1)

x2 = [1,2,3,4]

y2 = [4,3,2,1]

plt.plot(x2,y2)

plt.xlabel('x – axis', fontsize=16)

plt.ylabel('y – axis', fontsize=16)

plt.title('Two lines on same graph!')

plt.show()
```

Data plotting - Python

```
import matplotlib.pyplot as plt

x1 = [1,2,3,4]

y1 = [1,2,3,4]

plt.plot(x1, y1, label = "line 1")

x2 = [1,2,3,4]

y2 = [4,3,2,1]

plt.plot(x2,y2, label = "line 2")

plt.xlabel('x – axis')

plt.ylabel('y – axis')

plt.title('Two lines on same graph!')

plt.legend()

plt.show()
```

Data plotting - Python

```
import matplotlib.pyplot as plt

import numpy as np

y = [1,2,3,4]

abc = ['one', 'two', 'three', 'four']

plt.bar(np.arange(len(y)),y,tick_label = abc, width=0.4)

plt.show()
```

Data plotting - Python

```
import matplotlib.pyplot as plt

Import numpy as np

y1 = [1,2,3,4]

y2 = [1,2,3,4]

tick_label = ['one', 'two', 'three', 'four']

plt.bar(np.arange(len(y1))-0.2,y1,tick_label = tick_label, width=0.4)

plt.bar(np.arange(len(y2))+0.2,y2,tick_label = tick_label, width=0.4)

plt.title('my bar chart')

plt.legend(labels=['a','b'])

plt.show()
```

Data ploting - Python

```
import matplotlib.pyplot as plt

ages =[2,5,70,40,30,45,50,45,43,40,44,60,7,13,57,18,90,77,32,21,20,40]

range = (0, 100)

bins = 10

plt.hist(ages, bins, range)

plt.xlabel('age')

plt.ylabel('# of people')

plt.title('My histogram')

plt.show()
```

Data ploting - Python

```
import matplotlib.pyplot as plt

ages =[2,5,70,40,30,45,50,45,43,40,44,60,7,13,57,18,90,77,32,21,20,40]

range = (0, 100)

bins = 10

plt.hist(ages, bins, range, edgecolor='black')

plt.xlabel('age')

plt.ylabel('# of people')

plt.title('My histogram')

plt.show()
```

Data ploting - Python

```
import matplotlib.pyplot as plt

activities = ['eat', 'sleep', 'work', 'play']

slices = [3, 7, 8, 6]

colors = ['r', 'y', 'g', 'b']

plt.pie(slices, labels = activities, colors=colors)

plt.show()
```

Data ploting - Python

```
import matplotlib.pyplot as plt

activities = ['eat', 'sleep', 'work', 'play']

slices = [3, 7, 8, 6]

colors = ['r', 'y', 'g', 'b']

plt.pie(slices, labels = activities, colors=colors,autopct = '%1.2f%%')

plt.show()
```

Data ploting - Python

```
import matplotlib.pyplot as plt

Import numpy as np

x = [1,2,3,4,5,6,7,8,9,10]

y = [2,4,5,7,6,8,9,11,12,12]

plt.scatter(x, y)

plt.xlabel('x – axis')

plt.ylabel('y – axis')

plt.title('scatter plot')

plt.xticks(np.arange(min(x), max(x),1.0))

plt.show()
```

Data ploting - Python

```
import matplotlib.pyplot as plt

import numpy as np

feature_x = np.linspace(0, 3.0, 30)

feature_y = np.linspace(0, 3.0, 30)

[X, Y] = np.meshgrid(feature_x, feature_y)

Z = X ** 2 + Y ** 2

plt.contourf(X,Y,Z)

plt.show()
```

Data ploting - Python

```
import matplotlib.pyplot as plt
import numpy as np
from mpl_toolkits import mplot3d
plt.axes(projection="3d")
x=[1,2,3,4,5]
y=[3,5,2,1,4]
z=[5,2,1,4,3]
plt.plot(x,y,z)
plt.show()
```